Tested Toolchain Guide

Pin these versions for the batch. Update this file when you re-test.

Last verified profile: Python Full Stack batch kit (macOS + Windows learners)

Required versions

Tool	Version target	Check command
Python	**3.11 or 3.12** (prefer 3.12)	`python3 --version`
pip	bundled with Python	`python3 -m pip --version`
Git	2.40+	`git --version`
MySQL Server	**8.0+**	`mysql --version`
Node.js	**20 LTS** (or 22 LTS)	`node --version`
npm	comes with Node	`npm --version`
Django	**5.x** (`django>=5.0,<6.0`)	`python -c "import django; print(django.get_version())"`
djangorestframework	current stable for Django 5	in `requirements.txt`
Vite / React	Vite 5/6 + React 18+	`npm create vite@latest`
VS Code	latest stable	Help → About

Python packages (typical project)

```
django>=5.0,<6.0
djangorestframework
mysqlclient
# or PyMySQL if mysqlclient build fails on Windows
python-dotenv
whitenoise
gunicorn
requests
pytest
```

React app: create with Vite; add `axios` only if the batch prefers it over `fetch`.

VS Code extensions

- Python (Microsoft)
- Pylance
- Django (optional)
- ESLint / Prettier (React weeks)
- MySQL (optional GUI) — Workbench is fine without it
- GitLens (optional)

Install cheat sheet

macOS (Homebrew)

```
brew install python@3.12 git mysql node@20
brew services start mysql
```

Windows

1. python.org installer — tick **Add to PATH**
2. Git for Windows
3. MySQL Installer (Server + Workbench)
4. Node LTS MSI

If `mysqlclient` fails: `pip install PyMySQL` and add `import PyMySQL; PyMySQL.install_as_MySQLdb()` in Django `__init__.py` (document in project README).

Known fixes

Symptom	Fix
`python` not found on Mac	Use `python3`
Django MySQL encode errors	Install/upgrade `mysqlclient` or use PyMySQL bridge
`node` / `npm` missing	Reinstall Node LTS; restart terminal
Port 8000 in use	`python manage.py runserver 8001`
Vite CORS to Django	`django-cors-headers` + `CORS_ALLOWED_ORIGINS`
WhiteNoise 404 static	`collectstatic` + middleware order after SecurityMiddleware

Day-0 checklist

- [ ] Paste version outputs (`python3`, `git`, `mysql`, `node`) in the shared sheet
- [ ] One sample Django+MySQL project migrates on your laptop
- [ ] One Vite app starts with `npm run dev`
- [ ] GitHub account + SSH/HTTPS push works

Lab assets assume this toolchain: `15_Lab_Assets/`.